

List of Assignments (Jan-Apr 2026)

Subject – Operating Systems Lab
Class – SE(Computer Engineering)

Course Code : 2114115
Sem : IV

1. . Explore the internal commands of linux like

LO2,3

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|---|-----------|-----------|
| 1. ls | 2. Mkdir | 3. chdir |
| 4. rmdir | 5. Cat | 6. rm |
| 7. mv | 8. Cp | 9. head |
| 10. tail | 11. Sort | 12. wc |
| 13. chown | 14. Chmod | 15. chgrp |
| 16. umask | 17. Ps | 18. pipe |
| 19. Redirection operators | | |
| 20. Explore commands for following: | | |
| a. Display top 10 processes in descending order, | | |
| b. Display the process with highest memory usage, | | |
| c. Display current user logged in and logname, | | |
| d. Display current shell, home directory, operating system type, current path setting, current working directory, | | |
| e. Display OS version, release number, kernel version. | | |

2. Write Shell Programs to

LO2,3

1. Add 2 numbers
2. Check if a number entered is even and odd
3. Find sum of n numbers
4. Determine if a person is eligible to vote or not
5. Display all filenames beginning with character 'a' and displays its contents
6. Find Factorial of a number
7. Check validity of a username and password with a function defined in the code

3. Write a C program to simulate **Non Pre-emptive Priority Scheduling Policy** and **Preemptive SJF scheduling** policy **LO1,4**
4. Write a C program to simulate the **Best Fit and Worst Fit** Memory Allocation Technique **LO1,4**
5. Write a C Program to simulate following **LRU and Optimal** Page Replacement techniques **LO1,4**
6. Write a C Program to simulate **Paging** technique **LO1,4**
7. Write a C Program to implement the **SSTF Disk Scheduling Policy** **LO1,4**
8. Write a C Program to implement the following **SCAN Disk Scheduling Policy** **LO1,4**
9. Write a C program to implement solution of **Producer consumer problem** through **Semaphore.** **LO1,4**
10. Write a C program to demonstrate the concept of **deadlock avoidance** through **Banker's Algorithm** **LO1,4**

Written Assignments:

1. **System Boot Process and OS Initialization: Research and document the system boot process on two different platforms: Windows and Linux.** **CO1**
2. **1) Compare Open-Source Operating Systems and Proprietary Operating Systems with respect to architecture, licensing, cost, security, customization, and real-world applications. Support your answer with suitable examples.**

2) Compare Distributed Operating Systems, Network Operating Systems, Cloud Operating Systems, Real-Time Operating Systems, and Mobile Operating Systems based on their architecture, key functions, performance requirements, scalability, and application domains. **CO6**